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Quadratic Sieve

$$x^2 \equiv N \mod p$$

$$x \equiv a_p \mod p \text{ or } x \equiv b_p \mod p$$

$$x^2 - 227179 \equiv 0 \mod 5$$

$$x \equiv 2, 3 \mod 5$$

$$(\langle 5 \rangle + 2)^2 - N = \langle 5 \rangle$$

```
$ sage ch11-quadratic-sieve.sage
---  -----  --------------------  ------------------------------
470  -6279  -1 * 3 * 7 * 13 * 23  [1, 0, 1, 0, 1, 0, 1, 0, 0, 1]
473  -3450  -1 * 2 * 3 * 5^2 * 23  [1, 1, 1, 0, 0, 0, 0, 0, 0, 1]
477    350  2 * 5^2 * 7            [0, 1, 0, 0, 1, 0, 0, 0, 0, 0]
482   5145  3 * 5 * 7^3            [0, 0, 1, 1, 1, 0, 0, 0, 0, 0]
493  15870  2 * 3 * 5 * 23^2       [0, 1, 1, 1, 0, 0, 0, 0, 0, 0]
---  -----  --------------------  ------------------------------
212460² ≡ 169050² (mod 227179)
227179 = 157 × 1447
```

So we see 5 divides all x that are of the form $5a + 2$ or $5a + 3$.

Exercise 11.4: factorise 1679

We are given $(a, b) = (-1, 2), (5, 4)$ and $1679 = 41^2 - 2$.

$$\mathbb{Z}[\sqrt{2}] \to \mathbb{Z}/\langle 1679 \rangle$$

$$a + b\sqrt{2} \to a + 41b$$

$$N((-1, 2)) = N((5, 4)) = -7$$

$$\phi((-1, 2)) = 81 = 3^4, \qquad N((5, 4)) = 169 = 13^2$$

$$\begin{aligned}(-1 + 2\sqrt{2})(5 + 4\sqrt{2}) &= 11 + 6\sqrt{2} \\ &= (3 + \sqrt{2})^2\end{aligned}$$

$$\phi(3 + \sqrt{2}) = 44$$

$$\Rightarrow 44^2 = (3^2 13)^2 \mod 1679$$

```
sage: var("x")
x
sage: R.<a> = NumberField(x^2 - 2)
sage: sqrt((-1 + 2*a)*(5 + 4*a))
a + 3
sage: (3 + a)^2
6*a + 11
sage: 6*41 + 11
257
sage: Mod(44, 1679)^2 == 257^2
False
sage: Mod(44, 1679)^2
257
sage: 3^2 * 13
117
sage: Mod(44, 1679)^2 == 117^2
True

sage: gcd(1679, 117 + 44)
23
sage: gcd(1679, 117 - 44)
73
sage: 23 * 73
1679
```

$a^2 - 6b^2$ is divisible by 7 means 6 is a square modulo 7

1. $(c^2)^{-1} = (c^{-1})^2$ so we see the inverse of a square is also a square.
2. $a^2 - 6b^2 \equiv 0 \mod 7 \Rightarrow a^2 b^{-2} \equiv 6 \mod 7$
3. `kronecker(6, 7) = -1`, so 7 cannot be a divisor of the norm.

By the same argument, we can see that 6 modulo p must be a quadratic residue.

Prime Ideal $\mathfrak{p} = \langle p, \sqrt{d} - r \rangle$

We saw in chapter 5 that

$$\mathbb{Z}_K/\mathfrak{p}_i \cong \mathbb{F}_p[X]/\langle \bar{g}_i(X) \rangle$$

so the quotient ring contains $\mathbb{F}_p$.

We know $\mathbb{Z}_K/\mathfrak{p}$ is a finite field with a cardinality measured by the norm which is a power of p. All finite fields contain a subfield $\mathbb{F}_p$ by Cauchy. In this subfield p is the zero.

Since $p \in \mathfrak{p}$, we see that $\phi(\mathbb{Z}) = \mathbb{F}_p$, which is the restriction of $\phi|_{\mathbb{Z}}$.

1. The ideal $\mathfrak{p}$ is a factorization of $\langle a + b\sqrt{d} \rangle$, where $N(\langle a + b\sqrt{d} \rangle) = a^2 - db^2$.
2. We assume $p | N(\langle a + b\sqrt{d} \rangle)$, which means $p \in \langle a + b\sqrt{d} \rangle \subseteq \mathfrak{p}$.
3. Consider the map ϕ which is a homomorphism.
4. We see that $\phi(a + b\sqrt{d}) = \mathfrak{p}$. Rearranging this, we get $\phi(\sqrt{d}) = -\phi(ab^{-1})$.
5. We showed in `ch9.md` (title $\mathbb{Z}_K = \mathbb{Z} + \pi\mathbb{Z}_K$) that the cosets of $\mathbb{Z}_K/\mathfrak{p}$ are of the form $a + \mathfrak{p}$ where $a \in \{0, ..., p-1\}$.
6. Finally we have $a + b\sqrt{d} = pq + (a - pq) + b\sqrt{d}$, where $|a - pq| < p$. Then we can set $r = ab^{-1}$, and we see $\mathfrak{p} = \langle p, \sqrt{d} - r \rangle$.
7. Finally we have $a + b\sqrt{d} = b(ab^{-1} + \sqrt{d})$, and minus some multiple of p, so $r \equiv -ab^{-1} \mod p$.

$$\mathfrak{p} = \langle p, \sqrt{d} - r \rangle$$

$\mathfrak{p}$ has norm p due to its coset representation, and the right hand side also has norm p due to how we constructed it.

Restriction of ϕ to $\mathbb{Z}$ is $\mathbb{F}_p$

Let a, a' be such that $a \equiv a' \mod p \Rightarrow a \equiv a' \mod \mathfrak{p}$ since $p \in \mathfrak{p}$.

Likewise $a \equiv a' \mod \mathfrak{p} \Rightarrow \langle a - a' \rangle \subseteq \mathfrak{p}$ but $N(\mathfrak{p}) = p^k \Rightarrow N(\langle a - a' \rangle) | p^k$ so $p | a - a' \Rightarrow a \equiv a' \mod p$.

So the cosets of $a + \mathfrak{p}$ with $a \in \{0, ..., p-1\}$ are distinct.

All Cases

1. If $p > 2, \left(\frac{d}{p}\right) = 1$, or $p = 2, d \equiv 1 \mod 8$ then
$$\langle p \rangle = \mathfrak{p}_1 \mathfrak{p}_2$$
2. If $p > 2, p | d$, or $p = 2, d \equiv 2, 3 \mod 4$ then
$$\langle p \rangle = \mathfrak{p}^2$$
3. If $p > 2, \left(\frac{d}{p}\right) = 1$, or $p = 2, d \equiv 5 \mod 8$ then $\langle p \rangle$ is a prime ideal of $\mathbb{Z}_K$.

$\left(\frac{r}{p}\right) = 1 \Rightarrow \langle p \rangle = \mathfrak{p}_1 \mathfrak{p}_2$

Let $\mathfrak{p}_1 = \langle p, r + \sqrt{d} \rangle, \mathfrak{p}_2 = \langle p, r - \sqrt{d} \rangle$.

We prove first $\mathfrak{p}_1 \neq \mathfrak{p}_2$. Suppose $\mathfrak{p}_1 = \mathfrak{p}_2$, then $2a = (r + \sqrt{d}) + (r - \sqrt{d}) \in \mathfrak{p}_1$. But $2a \in \mathbb{Z}$ so $2a \in \mathfrak{p}_1 \cap \mathbb{Z} = \langle p \rangle$. Hence $p | 2a$ but this is impossible since p is odd.

Now multiply $\mathfrak{p}_1 \mathfrak{p}_2$

$$\begin{aligned} \mathfrak{p}_1 \mathfrak{p}_2 &= \langle p, r + \sqrt{d} \rangle \langle p, r - \sqrt{d} \rangle \\ &= \langle p \rangle I \end{aligned}$$

$$I = \langle p, r + \sqrt{d}, r - \sqrt{d}, (r^2 - d)/p \rangle$$

Since $\gcd(2r, p) = 1$, there are integers x, y such that

$$xp + y(2r) = 1$$

$$\Rightarrow 1 = xp + y(2r) = xp + (r + \sqrt{d}) + (r - \sqrt{d}) \in I$$

$$I = \langle 1 \rangle$$

$$\langle p \rangle = \mathfrak{p}_1 \mathfrak{p}_2$$